DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 9: Two-Fund Separation & the Capital Market Line

The Risk-Free Asset · Tangency Portfolio · Separation

August 12, 2026

Today's Agenda

1. Adding a risk-free asset

Today's Agenda

1. Adding a risk-free asset
2. The Capital Allocation Line and Sharpe ratio

Today's Agenda

1. Adding a risk-free asset
2. The Capital Allocation Line and Sharpe ratio
3. The tangency portfolio and the CML

Today's Agenda

1. Adding a risk-free asset
2. The Capital Allocation Line and Sharpe ratio
3. The tangency portfolio and the CML
4. The Two-Fund Separation Theorem

Learning Objectives

- Explain how a risk-free asset turns the frontier into a straight line.

Learning Objectives

- Explain how a risk-free asset turns the frontier into a straight line.
- Define the CAL and Sharpe ratio; identify the tangency portfolio.

Learning Objectives

- Explain how a risk-free asset turns the frontier into a straight line.
- Define the CAL and Sharpe ratio; identify the tangency portfolio.
- State and interpret the Two-Fund Separation Theorem.

Learning Objectives

- Explain how a risk-free asset turns the frontier into a straight line.
- Define the CAL and Sharpe ratio; identify the tangency portfolio.
- State and interpret the Two-Fund Separation Theorem.
- Explain why the CML dominates the risky-only frontier.

A Perfectly Flat, Zero-Risk Anchor

Tying a String to the Bullet

Every risky portfolio lives on the curved Markowitz bullet.

A Perfectly Flat, Zero-Risk Anchor

Tying a String to the Bullet

Every risky portfolio lives on the curved Markowitz bullet. The risk-free asset is one point at $\sigma = 0$.

A Perfectly Flat, Zero-Risk Anchor

Tying a String to the Bullet

Every risky portfolio lives on the curved Markowitz bullet. The risk-free asset is one point at $\sigma = 0$. Tie a taut string from that anchor to any risky portfolio P, pull it straight – every point on the string is achievable.

A Perfectly Flat, Zero-Risk Anchor

Tying a String to the Bullet

Every risky portfolio lives on the curved Markowitz bullet. The risk-free asset is one point at $\sigma = 0$. Tie a taut string from that anchor to any risky portfolio P, pull it straight – every point on the string is achievable. Tie it to just the *right* P, and the straight line beats the curve almost everywhere.

The Capital Allocation Line

Definition

Fraction y in risky portfolio P, $1 - y$ risk-free:

$$\mu_C = r_f + y(\mu_P - r_f), \qquad \sigma_C = y\,\sigma_P$$

The Capital Allocation Line

Definition

Fraction y in risky portfolio P, $1 - y$ risk-free:

$$\mu_C = r_f + y(\mu_P - r_f), \qquad \sigma_C = y\,\sigma_P$$

Eliminating y:

$$\mu_C = r_f + \left(\frac{\mu_P - r_f}{\sigma_P}\right)\sigma_C$$

The Capital Allocation Line

Definition

Fraction y in risky portfolio P, $1 - y$ risk-free:

$$\mu_C = r_f + y(\mu_P - r_f), \qquad \sigma_C = y\,\sigma_P$$

Eliminating y:

$$\mu_C = r_f + \left(\frac{\mu_P - r_f}{\sigma_P}\right)\sigma_C$$

Linear in σ_C – a straight line, unlike the curved risky-only frontier.

The Sharpe Ratio

Definition

The CAL's slope, $\dfrac{\mu_P - r_f}{\sigma_P}$, is the **Sharpe ratio**: excess return per unit of risk.

The Sharpe Ratio

Definition

The CAL's slope, $\dfrac{\mu_P - r_f}{\sigma_P}$, is the **Sharpe ratio**: excess return per unit of risk.

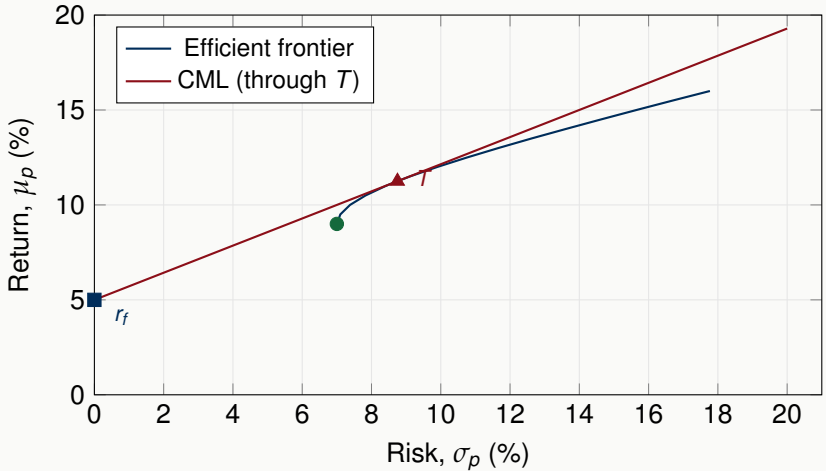
Every risky portfolio defines its own CAL, with its own slope.

The Tangency Portfolio and the CML

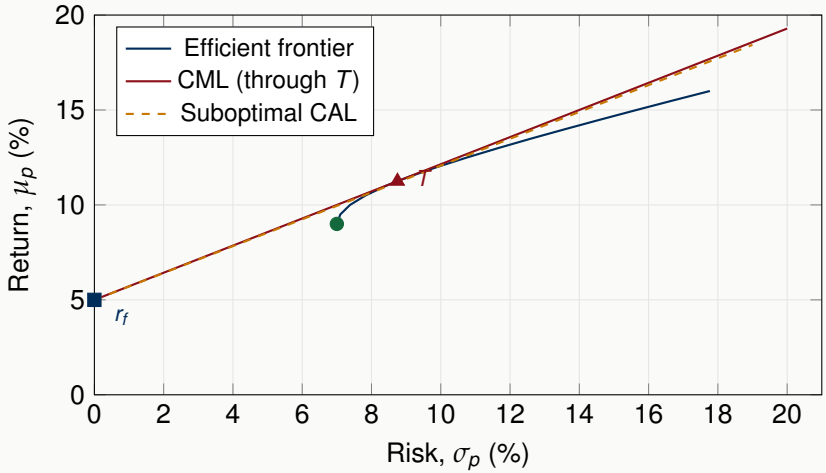
Definition

T = the risky portfolio with the **highest** Sharpe ratio – where a line from $(0, r_f)$ is exactly tangent to the frontier. Its CAL is the **Capital Market Line**.

CML vs. the Risky Frontier



CML vs. the Risky Frontier



Comparing Sharpe Ratios

$r_f = 5\%$

- $T = (8.75\%, 11.25\%)$: Sharpe $= (11.25 - 5)/8.75 = \mathbf{0.7143}$

Comparing Sharpe Ratios

$r_f = 5\%$

- $T = (8.75\%, 11.25\%)$: Sharpe $= (11.25 - 5)/8.75 = \mathbf{0.7143}$
- $(9.9\%, 12\%)$: Sharpe $= (12 - 5)/9.9 = 0.7071$

Comparing Sharpe Ratios

$r_f = 5\%$

- $T = (8.75\%, 11.25\%)$: Sharpe $= (11.25 - 5)/8.75 = \mathbf{0.7143}$
- $(9.9\%, 12\%)$: Sharpe $= (12 - 5)/9.9 = 0.7071$
- $(11.67\%, 13\%)$: Sharpe $= (13 - 5)/11.67 = 0.6855$

Comparing Sharpe Ratios

$r_f = 5\%$

- $T = (8.75\%, 11.25\%)$: Sharpe $= (11.25 - 5)/8.75 = \mathbf{0.7143}$
- $(9.9\%, 12\%)$: Sharpe $= (12 - 5)/9.9 = 0.7071$
- $(11.67\%, 13\%)$: Sharpe $= (13 - 5)/11.67 = 0.6855$

Comparing Sharpe Ratios

$r_f = 5\%$

- $T = (8.75\%, 11.25\%)$: Sharpe $= (11.25 - 5)/8.75 = \mathbf{0.7143}$
- $(9.9\%, 12\%)$: Sharpe $= (12 - 5)/9.9 = 0.7071$
- $(11.67\%, 13\%)$: Sharpe $= (13 - 5)/11.67 = 0.6855$

T wins – exactly as tangency requires.

The Two-Fund Separation Theorem

Two-Fund Separation

Every investor holds a combination of exactly **two** funds:

- the risk-free asset, and
- the **same** tangency portfolio T, common to everyone.

Investors differ *only* in the split y vs. $1-y$ – never in the composition of the risky fund.

One Recipe, Many Portion Sizes

The Restaurant's "Special"

The kitchen has found the single best dish physically possible – "the special."

One Recipe, Many Portion Sizes

The Restaurant's "Special"

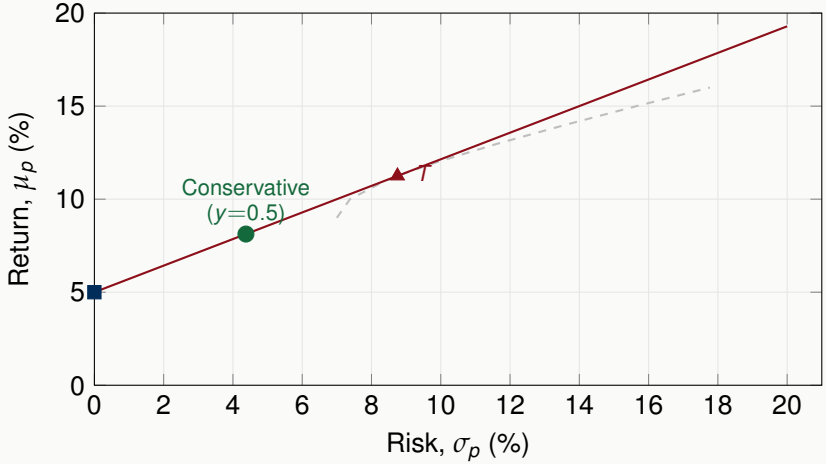
The kitchen has found the single best dish physically possible – "the special." Every customer, whatever their appetite for risk, orders the *same* special, just in a different portion, topped up with plain rice (risk-free asset).

One Recipe, Many Portion Sizes

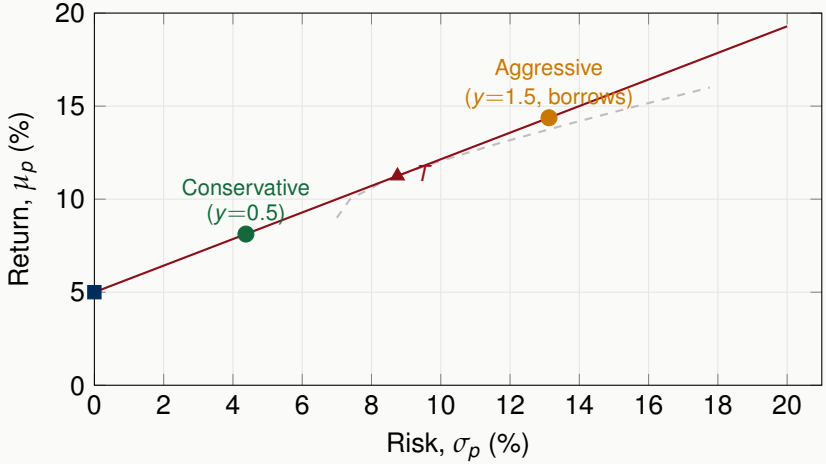
The Restaurant's "Special"

The kitchen has found the single best dish physically possible – "the special." Every customer, whatever their appetite for risk, orders the *same* special, just in a different portion, topped up with plain rice (risk-free asset). No one asks for a different recipe – only a different portion size.

Two Investors, One Line



Two Investors, One Line



Where Does Leverage Come From?

Remark

$y > 1$: borrow at r_f to invest *more* than 100% of capital in T – a leveraged position.

Where Does Leverage Come From?

Remark

$y > 1$: borrow at r_f to invest *more* than 100% of capital in T – a leveraged position.
$y < 0$: short T, invest more than 100% in the risk-free asset.

Where Does Leverage Come From?

Remark

$y > 1$: borrow at r_f to invest *more* than 100% of capital in T – a leveraged position.
$y < 0$: short T, invest more than 100% in the risk-free asset. Both lie on the same CML.

Why Does Separation Happen?

The Logic

The CML dominates the risky-only frontier everywhere except at T.

Why Does Separation Happen?

The Logic

The CML dominates the risky-only frontier everywhere except at T. So any investor with access to the risk-free asset does strictly better on (r_f, T) than on any other risky mix.

Why Does Separation Happen?

The Logic

The CML dominates the risky-only frontier everywhere except at T. So any investor with access to the risk-free asset does strictly better on (r_f, T) than on any other risky mix. The *only* remaining choice is y – personal risk aversion (Week 2's γ or a) affects only *position* on the CML, never the risky recipe.

Key Takeaways

Key Takeaway

- A risk-free asset turns the achievable set into a straight line: the **CAL**.

Key Takeaways

Key Takeaway

- A risk-free asset turns the achievable set into a straight line: the **CAL**.
- The **tangency portfolio** T maximizes the Sharpe ratio; its CAL is the **CML**.

Key Takeaways

Key Takeaway

- A risk-free asset turns the achievable set into a straight line: the **CAL**.
- The **tangency portfolio** T maximizes the Sharpe ratio; its CAL is the **CML**.
- **Two-Fund Separation**: every investor holds only r_f and T, differing only in the split.

Key Takeaways

Key Takeaway

- A risk-free asset turns the achievable set into a straight line: the **CAL**.
- The **tangency portfolio** T maximizes the Sharpe ratio; its CAL is the **CML**.
- **Two-Fund Separation**: every investor holds only r_f and T, differing only in the split.
- This is the conceptual doorway to a single, common market portfolio – and CAPM.

Check Your Understanding

Discuss (2 minutes)

1. $r_f = 6\%$, Q: μ_Q=14%, σ_Q=16%. Sharpe ratio? Is Q necessarily T?

Check Your Understanding

Discuss (2 minutes)

1. $r_f = 6\%$, Q: μ_Q=14%, σ_Q=16%. Sharpe ratio? Is Q necessarily T?
2. $r_f = 5\%$, $T = (8.75\%, 11.25\%)$. Find y for $\sigma_C = 12\%$, and μ_C.

Check Your Understanding

Discuss (2 minutes)

1. $r_f = 6\%$, Q: μ_Q=14%, σ_Q=16%. Sharpe ratio? Is Q necessarily T?
2. $r_f = 5\%$, $T = (8.75\%, 11.25\%)$. Find y for $\sigma_C = 12\%$, and μ_C.
3. Why do investors with very different risk aversion hold risky assets in *identical* relative proportions?

Looking Ahead

If every investor holds the same T, and the market is the sum of all holdings...

Looking Ahead

If every investor holds the same T, and the market is the sum of all holdings...

...then in equilibrium, T **must be** the market portfolio.

Looking Ahead

If every investor holds the same T, and the market is the sum of all holdings...

...then in equilibrium, T **must be** the market portfolio.

CAPM – Week 4